
RL Team Project Proposal : Safe-GTA: Physics-Informed Generative Trajectory Augmentation for Safe Offline RL

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1 Project Overview

1.1 Profile

Track: 1. Algorithm

Abstract and project goal: This project addresses the critical issues of data quality and safety in offline reinforcement learning (RL) for autonomous driving. Standard offline datasets often suffer from data scarcity and sub-optimality, containing dangerous trajectories that hinder policy convergence. We propose **Safe-GTA**, a framework that leverages Diffusion Models with Physics-Informed Safety Guidance to refine sub-optimal data and generate high-quality, safe trajectory augmentations. By integrating a Safety Critic into the denoising process, our method ensures that generated samples strictly adhere to track boundaries and kinematic constraints. The ultimate goal is to provide a robust data augmentation pipeline that improves the performance and safety of offline agents in out-of-distribution scenarios.

TL;DR: Safe-GTA utilizes Diffusion Models and Safety Critic gradients to repair and augment offline driving data, ensuring zero-constraint violation trajectories for safer offline RL training.

1.2 Motivation

Why is the problem interesting? Autonomous driving is a safety-critical application where traditional RL's trial-and-error approach is prohibited in the real world. Offline RL offers a solution by learning from static datasets, but the "Distribution Shift" problem often leads to catastrophic collisions during evaluation. Developing a method that can autonomously improve dataset quality while guaranteeing physical safety boundaries is essential for deploying RL in real-world transportation systems.

- **Challenge 1: Balancing Safety and Reward.** Purely reward-driven generative models often produce "high-reward but risky" behaviors, such as cutting corners or exceeding physical limits.
- **Challenge 2: Temporal and Kinematic Consistency.** Generating long-horizon trajectories that are both smooth and physically plausible is difficult, as small errors in action-level generation accumulate over time.
- **Challenge 3: Handling Sub-optimal Data.** Transforming "noisy" or dangerous demonstrations into expert-level trajectories without losing diversity remains a significant technical hurdle.

Justify why the problem remains open or unsolved? While generative models like Normalizing Flows and Diffusion Models have been applied to RL, most existing works focus on online planning

or simple action-level constraints. The problem remains open because current methods often incur high inference latency or fail to ensure long-term physical consistency in complex driving tracks.

Relevant prior works include seminal papers such as **CQL** Kumar et al. [2020] and **Diffuser** Janner et al. [2022]. In the last two years, recent advancements like **FISOR** Do et al. [2024] (2024), **GTA** Zheng et al. [2024] (2024), and **FlowPG** Brahmanage et al. [2023] (2023) have explored feasibility guidance and generative layers. However, a unified framework that combines trajectory-level repair with dynamic safety budgeting (as proposed in **TREBI** Qin et al. [2023]) is still missing.

State-of-the-Art (SOTA) methods: In our opinion, **FISOR** Do et al. [2024] and **FlowPG** Brahmanage et al. [2023] represent the current SOTA in action-constrained and feasibility-guided RL. However, **Safe-GTA** aims to surpass these by focusing on offline data augmentation rather than just online policy constraints.

2 Related Work

Existing research in Safe Reinforcement Learning (RL) has explored various mechanisms to enforce constraints, ranging from real-time optimization to generative modeling.

Feasibility-guided Diffusion (FISOR): FISOR [Do et al., 2024] introduces a feasibility-guided mechanism that treats safety constraints as an energy-guided function to push the denoising process toward a safe region. While effective, FISOR is designed as an online execution policy, which leads to high inference latency in real-time autonomous driving. In contrast, **Safe-GTA** acts as an offline data augmentation tool, repairing the dataset before the training phase begins.

Trajectory Planning via Diffusion (SDD Planner): SDD Planner [Huang et al., 2024] models driving styles and safety as conditional constraints for online trajectory planning. However, its performance is heavily dependent on the quality of pre-trained models. Our work addresses the core issue of data sub-optimality in offline datasets by utilizing dynamic safety budgeting [Qin et al., 2023] to generate high-quality, diverse safe samples.

Safe Policy Improvement (SPI): Recent works like SPI [Yang et al., 2025] focus on action-level safety by filtering high-risk actions using a safety value function. However, action-level filtering often ignores temporal and kinematic consistency, leading to jittery vehicle control. **Safe-GTA** operates at the trajectory-level, ensuring that the generated paths are smooth and physically plausible over a long horizon.

Generative Trajectory Augmentation (GTA): While the original GTA [Zheng et al., 2024] framework successfully alleviates data scarcity in offline RL, it lacks explicit safety guarantees. Our proposed framework extends this by incorporating physics-informed gradients to ensure all augmented data points strictly adhere to environment boundaries.

3 Problem Formulation

We consider a safety-constrained Markov Decision Process (CMDP) defined by the tuple $\mathcal{M} := (\mathcal{S}, \mathcal{A}, \mathcal{P}, \gamma, r, \mathcal{C})$. Here, $\mathcal{S}$ and $\mathcal{A}$ represent the state and action spaces, $\mathcal{P} : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ denotes the transition kernel, $\gamma \in (0, 1)$ is the discount factor, and $r : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ is the bounded reward function, and $\mathcal{C} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}_{\geq 0}$ is the safety cost function indicating constraint violations (e.g., track boundaries). The RL learner is provided with an offline dataset $\mathcal{D} = \{(s_i, a_i, r_i, s'_i, c_i)\}_{i=1}^N$ containing a mixture of expert and sub-optimal demonstrations, where $c_i = \mathcal{C}(s_i, a_i)$.

The optimization problem is divided into two phases: trajectory augmentation and policy learning. First, we aim to learn a conditional diffusion model to generate augmented trajectories $\tau = \{(s_0, a_0), \dots, (s_T, a_T)\}$ that maximize the expected return $J(\tau) = \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right]$ while minimizing the cumulative safety cost.

To distinguish from the environmental timestep t , we denote the diffusion generation step as k . Let $x \equiv \tau$ represent the trajectory variable in the diffusion space. The augmentation is achieved by

solving the reverse diffusion process:

$$x^{k-1} = \text{Denoise}(x^k) + \alpha \nabla_x \mathcal{J}_R(x^k) - \beta \nabla_x \mathcal{J}_C(x^k)$$

where x^k represents the noisy trajectory at diffusion step k , α is the reward guidance scale, $\mathcal{J}_R$ is the reward predictor (estimating $J(\tau)$), $\mathcal{J}_C$ is the safety critic (estimating cumulative cost), and β is a dynamic safety budget parameter controlling the risk tolerance.

Second, we utilize the augmented safe dataset to optimize a conservative policy π^* that maximizes the expected return $V^\pi(s)$ while satisfying the safety constraints. We assume that the environment kinematics are locally continuous and that the Safety Critic can accurately estimate the cost gradient $\nabla_x \mathcal{J}_C$ within the support of the offline data.

4 Empirical Evaluation

4.1 Performance Metrics

We evaluate the proposed Safe-GTA framework using the following key metrics:

- **Normalized Return:** We report the standard episodic return normalized against expert performance to evaluate the agent’s optimality.
- **Safety Success Rate:** The ratio of evaluation episodes completed without any constraint violations (e.g., collisions or lane departures).
- **Constraint Violation Cost:** The cumulative cost incurred during an episode, serving as a direct measure of how strictly the policy adheres to safety boundaries.
- **Trajectory Fidelity (New Metric):** We measure the Fréchet distance between generated trajectories and expert demonstrations. This is required to ensure that the diffusion-based "repair" process maintains realistic driving patterns while enforcing safety.

4.2 Baseline Methods

- **CQL [Kumar et al., 2020]:** A standard offline RL algorithm that addresses distribution shift via conservative Q-value estimation without explicit trajectory augmentation.
- **GTA [Zheng et al., 2024]:** A generative trajectory augmentation framework that uses diffusion models for data expansion but lacks physics-informed safety constraints.
- **FISOR [Do et al., 2024]:** A feasibility-guided diffusion model that serves as an online safety-constrained policy baseline.
- **Safe-GTA (Ours):** Our proposed method that combines diffusion-based repair with physics-informed safety gradients for offline data enhancement.

4.3 Visualization Analysis

To qualitatively demonstrate the effectiveness of the proposed safety guidance, we introduce two visualization techniques:

- **Trajectory Ghosting:** This technique presents multiple candidate paths simulated by the AI under various safety budgets β simultaneously, illustrating the risk-aware decision process.
- **Cost Heatmap:** This maps the distribution of generated state-action pairs over the environment to prove that the points are strictly confined within safe track regions.
- **Denoising Process Timelapse:** This visualization highlights the reverse diffusion process from pure noise ($k = K$) to a clean, safe trajectory ($k = 0$).

4.4 Benchmark Tasks or Datasets

- **MetaDrive Simulator [Li et al., 2022]:** A realistic physics engine featuring diverse, procedurally generated track geometries and complex traffic scenarios.

- **Offline Safe RL (OSRL) Benchmark [Liu et al., 2023]:** We utilize the official OSRL datasets, which provide various quality levels (expert, medium, and random trajectories) to evaluate the framework’s ability to "repair" and augment sub-optimal data.

5 Methodology

The high-level idea of Safe-GTA is to transform the offline reinforcement learning task into a constrained generative modeling problem, where a Diffusion Model serves as a "physics-informed repair engine" for sub-optimal data. Our solution decomposes the trajectory augmentation process into three sequential stages: (1) Conditional Denoising, where raw trajectories from the OSRL dataset are refined into smooth, expert-like paths through a reverse diffusion process; (2) Physics-Informed Guidance, where an external Safety Critic provides real-time gradients $\nabla_x \mathcal{J}_C$ to steer the denoising coordinates away from track boundaries; and (3) Dynamic Budgeting, which utilizes a tunable safety parameter β (inspired by TREBI Qin et al. [2023]) to generate a diverse spectrum of safe-to-aggressive trajectories. This "repair-before-training" workflow ensures that the downstream RL agent (e.g., CQL Kumar et al. [2020]) learns from a high-quality, zero-constraint-violation dataset, significantly mitigating the distribution shift and safety risks inherent in standard offline RL.

6 Experimental Results

6.1 Evaluation of Baseline Methods

We will evaluate the effectiveness of Safe-GTA by comparing it against several established baselines in the MetaDrive environment using the OSRL benchmark. The evaluation will focus on the following key aspects:

- **Comparative Performance:** We will compare our method against CQL Kumar et al. [2020], which represents a model without augmentation, and Vanilla GTA Zheng et al. [2024], which performs augmentation without safety constraints. This comparison is designed to highlight the specific contribution of our physics-informed guidance in reducing constraint violations.
- **Safety-Critical Efficiency:** We plan to record the Safety Success Rate and Normalized Return across different training stages. We hypothesize that Safe-GTA will maintain a significantly lower Constraint Violation Cost compared to Vanilla GTA, especially in high-density traffic scenarios where boundary adherence is critical.
- **Quantitative Fidelity:** To justify the use of diffusion models, we will report the Fréchet distance between generated trajectories and expert data. Safe-GTA is expected to achieve higher fidelity than action-level baselines like FISOR Do et al. [2024], proving that trajectory-level "repair" better preserves the underlying driving logic while ensuring zero-constraint violations.
- **Qualitative Visualization:** Beyond numerical metrics, we will provide three key visual demonstrations. First, **Trajectory Ghosting** will visualize the multi-modal safe paths simulated under different safety budgets β . Second, a **Denoising Process Timelapse** will highlight the reverse diffusion process from pure noise ($k = K$) to a clean, safe trajectory ($k = 0$). Additionally, **Cost Heatmaps** will be utilized to empirically demonstrate whether the state-action pairs generated by Safe-GTA are strictly confined within the feasible region $\mathcal{C}(s)$.

7 Expected Contributions of Each Team Member

Our team will collaborate through weekly synchronization meetings and use a shared code repository to ensure seamless integration of the Safe-GTA modules. The initial division of labor is as follows:

- **Chang Hao Yang:** Primarily responsible for the *Conditional Trajectory Denoising* stage. This includes designing and training the Diffusion Model on the OSRL dataset to ensure it can effectively "repair" noisy or sub-optimal trajectories into expert-level behaviors.

- **Bo Yu Shen:** Focuses on the *Physics-Informed Safety Guidance* module. This member will be responsible for pre-training the *Safety Critic* network and implementing the gradient-based update rule ($\beta \nabla S$) to steer the diffusion process away from track boundaries.
- **Yu-Chun Chen:** Responsible for *System Integration and Empirical Evaluation*. This includes setting up the MetaDrive simulator, implementing the *Conservative Q-Learning (CQL)* baseline, and developing visualization tools such as Trajectory Ghosting and Cost Heatmaps.

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